Hospital OS

Next-Generation AI-Powered Hospital Management System — Designed to streamline clinical workflows, enhance patient outcomes, and deliver operational intelligence.

6+	5	AI	100%
CORE MODULES	USER ROLES	POWERED TRIAGE	AUDIT COMPLIANT

CONFIDENTIAL • Product Overview • v2.0

01 — EXECUTIVE SUMMARY

Revolutionizing Hospital Operations

Hospital OS is a comprehensive, cloud-native hospital management platform that unifies every facet of clinical and operational workflows under a single intelligent system.

End-to-End Patient Lifecycle

From registration and AI-powered triage to consultation, lab diagnostics, pharmacy dispensing, and discharge — every touchpoint is digitized and connected.

AI-Powered Clinical Intelligence

Integrated GPT-4o triage engine provides instant risk stratification, differential diagnoses, and SOAP-format clinical summaries for every patient encounter.

Real-Time Executive Dashboard

Live KPIs, revenue analytics, bed occupancy tracking, patient flow trends, and inventory alerts — all auto-refreshing every 60 seconds.

HIPAA-Ready Compliance

Complete audit trail for every action, encrypted password storage, role-based access control, and comprehensive system logging across all modules.

Why Hospital OS?

Traditional hospital software is fragmented — separate systems for registration, billing, lab, and pharmacy that don't communicate. Hospital OS eliminates silos with a unified, real-time platform built on modern cloud architecture. The result: faster patient throughput, fewer errors, complete financial visibility, and clinical decision support powered by artificial intelligence.

02 — TECHNOLOGY PLATFORM

Built on a Modern, Scalable Stack

Enterprise-grade technologies chosen for performance, security, and maintainability.

⚡

Next.js 16

FRONTEND & API FRAMEWORK

⚛️

React 19

USER INTERFACE LIBRARY

📝

TypeScript 5

TYPE-SAFE DEVELOPMENT

🐘

PostgreSQL

RELATIONAL DATABASE

🔌

Prisma ORM

DATABASE TOOLKIT

☁️

Supabase

CLOUD INFRASTRUCTURE

🤖

OpenAI GPT-4o

AI TRIAGE ENGINE

🔗

n8n Workflows

AUTOMATION & INTEGRATIONS

🎨

Tailwind CSS v4

DESIGN SYSTEM

Architecture Highlights

Server-side rendering with Next.js Server Actions ensures lightning-fast page loads and secure data processing. Prisma ORM with connection pooling delivers efficient database

queries. Real-time data sync through path revalidation and polling keeps all dashboards current. Turbopack compilation provides sub-second development builds.

03 — CORE MODULES

Comprehensive Hospital Modules

Six purpose-built modules covering every hospital department and workflow.

R

Reception & Patient Registration

RECEPTIONIST

Streamlined patient intake with intelligent ID generation and department routing. The reception module serves as the entry point for all patient encounters.

- ✓ Full patient registration with demographics, contact details, and Aadhaar ID
- ✓ Auto-generated unique Patient ID (PAT-YYYYMMDD-XXXX format)
- ✓ Auto-generated Appointment ID with date-stamped tracking
- ✓ Department selection (General, Cardiology, Orthopedics, Pediatrics)
- ✓ Webhook-powered official ID generation via n8n automation
- ✓ Success confirmation modal with generated IDs

🧠 AI-Powered Clinical Triage

The crown jewel of Hospital OS — an intelligent triage engine that transforms patient intake

👤 **Patient Info**
Demographics & contact

→

🩺 **Symptoms**
24+ symptom selection

→

📋 **History & Vitals**
PMH, meds, allergies

→

🤖 **AI Assessment**
GPT-4o analysis

- **3-Level Risk Stratification:** Emergency (Red), Urgent (Amber), Routine (Green)
- **Diagnostic Recommendations:** 4-8 targeted tests per assessment
- **Smart Escalation:** Vitals-based (SpO2, HR, Temp) and age-based (pediatric/elderly) alerts
- **Differential Diagnoses:** 4-6 possible conditions with confidence levels
- **SOAP Clinical Summary:** Professional documentation (Subjective, Objective, Assessment, Plan)
- **Dual Engine:** OpenAI primary with rule-based fallback for 100% uptime guarantee

D

Doctor Clinical Console

DOCTOR

A powerful clinical workspace combining patient queue management, electronic health records, lab ordering, prescriptions, and admission management in a single split-view interface.

- ✓ Real-time patient queue with status tracking (Pending, In Progress, Admitted)
- ✓ Electronic Health Record (EHR) creation with diagnosis and clinical notes
- ✓ Digital prescription writing with pharmacy order creation
- ✓ Complete patient history timeline across all encounters
- ✓ Walk-in patient registration for unscheduled visits
- ✓ AI triage results viewer showing risk levels and clinical summaries
- ✓ One-click lab test ordering with automated barcode generation
- ✓ In-patient admission management with bed allocation
- ✓ Medical notes system (Admission Notes, Routine Checks, Nursing, Discharge Advice)
- ✓ Discharge summary generation with PDF export

L

Laboratory Management

LAB TECHNICIAN

Efficient lab worklist management with order tracking, result upload, and automated report generation through AI-powered analysis.

- ✓ Pending orders worklist with real-time status updates
- ✓ Completed tests archive with historical data
- ✓ Result value and remarks entry with file upload capability
- ✓ Barcode-based order tracking for specimen management
- ✓ Webhook integration for AI-powered result analysis and PDF generation
- ✓ Real-time statistics dashboard (pending count, completed today)
- ✓ Complete audit trail for every result upload
- ✓ Patient details display with doctor and test type information

P

Pharmacy & Billing

PHARMACIST

Complete pharmacy operations including inventory management, prescription fulfillment, point-of-sale billing, and stock monitoring with automated alerts.

- ✓ Medicine inventory with batch tracking (batch no, expiry, rack location)
- ✓ Smart search and add-to-cart POS billing system
- ✓ Doctor prescription order queue with one-click processing
- ✓ Automated stock deduction upon dispensing
- ✓ Low stock threshold alerts and expiry date warnings
- ✓ New batch and medicine registration with flexible pricing
- ✓ Invoice generation with PDF export via webhook
- ✓ Complete sales audit log with batch-level tracking

X

Discharge Management

ADMIN STAFF

Structured discharge workflow ensuring all clinical, financial, and administrative requirements are completed before patient release.

- ✓ Admitted patient list with admission duration and status tracking
- ✓ Discharge checklist (Documentation, Medications, Final Bill, Follow-up)
- ✓ Automated discharge summary generation in HTML and PDF formats
- ✓ Quick statistics for total admissions and ready-for-discharge patients
- ✓ Integration with billing module for final settlement
- ✓ Complete audit logging of all discharge events

A

Admin Intelligence Center

ADMINISTRATOR

Executive-level operational dashboard providing real-time visibility into hospital performance, finances, resources, and compliance.

- ✓ Live KPIs: patient count, admissions, lab orders, revenue
- ✓ Bed occupancy visualization with ward-level breakdown
- ✓ 7-day patient flow trend analysis
- ✓ Revenue breakdown by department (OPD, Lab, Pharmacy, IPD, Procedures)
- ✓ Inventory alerts for low stock and expiring medicines
- ✓ Live audit trail feed with color-coded action types
- ✓ Auto-refreshing every 60 seconds for real-time monitoring
- ✓ Appointment and scheduling overview

04 — PATIENT JOURNEY

Seamless End-to-End Workflow

Every patient touchpoint is digitized, tracked, and connected across departments.

1

Patient Arrival & Registration

Receptionist registers the patient with demographics and contact information. System auto-generates a unique Patient ID and Appointment ID. Department is assigned based on chief complaint.

2

AI-Powered Triage Assessment

Receptionist captures symptoms, medical history, and optional vitals. AI engine performs instant risk stratification, generates differential diagnoses, recommended tests, and a SOAP clinical summary. Emergency cases are immediately flagged.

3

Doctor Consultation & Clinical Decisions

Doctor reviews AI triage results, examines the patient, creates clinical notes in the EHR, orders lab tests, and writes prescriptions. Patient can be admitted as an in-patient if needed.

4

Laboratory Diagnostics

Lab technician receives the order with barcode tracking, performs tests, uploads results with remarks. AI-powered analysis generates a formatted report and notifies the ordering doctor.

5

Pharmacy Dispensing & Billing

Pharmacist receives prescription orders, verifies inventory, dispenses medicines with batch-level tracking, and generates invoices. Stock levels auto-update, low-stock alerts trigger automatically.

6

Discharge & Summary

Discharge checklist ensures all documentation, medications, billing, and follow-ups are completed. System generates a comprehensive discharge summary in PDF format for the patient's records.

05 — SECURITY & COMPLIANCE

Enterprise-Grade Security

Built from the ground up with healthcare compliance and data protection in mind.

🔒 Role-Based Access Control

Five distinct user roles (Admin, Doctor, Receptionist, Lab Technician, Pharmacist) with role-specific dashboards and permissions. Each user sees only what they need.

🔐 Encrypted Authentication

bcrypt-hashed passwords with salt rounds. Secure httpOnly session cookies. No plain-text credentials stored anywhere in the system.

📝 Complete Audit Trail

Every action logged with user ID, username, role, action type, module, entity, timestamp, and detailed JSON metadata. Full traceability for compliance audits.

🛡️ HIPAA-Ready Architecture

Designed with healthcare data protection principles. Secure data transmission, access logging, and audit capabilities aligned with regulatory requirements.

06 — INTEGRATIONS

Connected Ecosystem

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OpenAI GPT-4o

AI-powered clinical triage with dual-engine reliability

🔄

n8n Automation

6+ webhook workflows for registration, lab orders, result analysis, pharmacy,

☁️

Supabase Cloud

Managed PostgreSQL with connection pooling,

(AI primary, rule-based fallback).

and discharge PDF generation.

automatic backups, and horizontal scaling.

07 — DASHBOARD INTELLIGENCE

Real-Time Operational Visibility

Live insights that empower administrators to make data-driven decisions.

142	28	67	$48.5K
PATIENTS TODAY	ACTIVE ADMISSIONS	LAB ORDERS	TODAY'S REVENUE

📊 **Bed Occupancy Ring Chart**
Ward-level breakdown: ICU, General, Private — available, occupied, maintenance

📈 **7-Day Patient Flow Trend**
Daily registration volumes visualized for capacity planning

💰 **Revenue by Department**
OPD, Lab, Pharmacy, IPD, Procedures — horizontal bar breakdown

🚨 **Inventory & Audit Alerts**
Low stock warnings, expiring batches, and live activity feed

Auto-Refreshing Intelligence

The admin dashboard automatically refreshes every 60 seconds, ensuring administrators always see the latest operational data without manual intervention. From bed availability to revenue tracking, every metric updates in real time — empowering proactive hospital management.

08 — SUMMARY

Everything Your Hospital Needs

Complete Solution

Registration, triage, consultation, lab, pharmacy, billing, discharge, and admin reporting — all in one platform.

AI-First Approach

GPT-4o powered triage, automated lab report analysis, and intelligent clinical decision support.

Modern Architecture

Next.js 16, React 19, TypeScript, PostgreSQL, Prisma — built for performance and scalability.

Future-Ready

Planned AI co-pilot for doctors, enhanced compliance modules, and advanced analytics in upcoming phases.

Ready to Transform Hospital Operations?

Hospital OS is built to scale with your hospital — from a single department to a multi-facility healthcare network.

✔ Cloud-Native ✔ AI-Powered ✔ HIPAA-Ready

✔ Real-Time Analytics ✔ Role-Based Access ✔ Full Audit Trail

Hospital OS by Avani Enterprise

Version 2.0 • Confidential Product Overview

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